

PHISHING EMAIL DETECTION AND AWARENESS REPORT

To analyse email samples, identify phishing indicators, classify email risks, and develop security awareness guidelines that help users recognize and avoid phishing attacks.

Shrilee Shah
Cybersecurity Intern
Future Interns

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Executive Summary

Phishing attacks remain one of the most common cybersecurity threats faced by individuals and organizations. Attackers often use deceptive emails to trick users into revealing sensitive information, clicking malicious links, or downloading harmful attachments.

This report presents an analysis of selected email samples to identify phishing characteristics and assess their potential risk. The assessment focused on reviewing email headers, sender information, authentication records, message content, and embedded links.

Three email samples were analysed and classified based on observed indicators:

- Phishing Email – High Risk
- Phishing Email – High Risk
- Safe Email – Low Risk

The analysis identified common phishing techniques such as sender spoofing, domain impersonation, suspicious links, social engineering tactics, and email authentication failures. In addition to email analysis, this report provides practical recommendations and awareness guidelines to help users recognize and avoid phishing attacks.

Objective and Scope of assessment

Objective

The objective of this assessment was to analyse email samples and identify potential phishing threats using email content review, header analysis, and security indicator identification.

The assessment aimed to:

- Identify phishing indicators within email messages.
- Analyse sender information and email authentication records.
- Classify emails based on risk level.
- Explain phishing techniques in simple and understandable language.
- Develop awareness guidelines to help users recognize and avoid phishing attacks.

Scope of assessment

The assessment included:

- Analysis of publicly available and educational email samples.
- Review of sender information and email headers.
- Inspection of embedded links and domains.
- Identification of phishing indicators and social engineering techniques.
- Risk classification of analysed emails.

Methodology

1. Email Collection

Phishing and legitimate email samples were collected from educational sources and publicly available datasets.

2. Header Analysis

Email headers were reviewed to examine sender information, return-path details, and email authentication mechanisms such as SPF, DKIM, and DMARC.

3. Content Inspection

The subject line, sender details, message content, and embedded links were inspected for phishing indicators and social engineering techniques.

4. Risk Assessment

Each email was classified based on identified indicators and assigned an appropriate risk level.

5. Awareness Development

Best practices, prevention guidelines, and employee awareness recommendations were developed based on the findings.

Tools Used

Tool	Purpose
Google Message Header Analyzer	Email header inspection and authentication analysis
MXToolbox Email Header Analyzer	Email header verification and routing analysis
Web Browser	Safe inspection of domains, links, and email content
Microsoft Word / Canva	Documentation and report preparation

Analysis Criteria

The following indicators were used during assessment:

- Sender authenticity
- Domain legitimacy
- SPF validation
- DKIM validation
- DMARC validation
- Suspicious links
- Social engineering tactics
- Urgency or fear-based language
- Brand impersonation
- Email content consistency

Case Study 1 – MetroFax Fax Notification Email

1. Email Overview

Field	Details
Email Type	Phishing Email
Risk Classification	Phishing
Risk Level	High
Claimed Organization	MetroFax
Sender Address	attack@attacker.example.com
Subject	Monday, March 7 2022

2. Email Summary

The email claims that the recipient has received a new fax and encourages them to click a link to view the attached document. The message impersonates MetroFax and attempts to create urgency by suggesting that a document is waiting to be reviewed.

3. Phishing Indicators Identified

a. Suspicious Sender Address

The sender email address uses an untrusted domain and does not belong to MetroFax.

Indicator: Sender spoofing / suspicious sender

b. Suspicious Link

The email contains a link directing users to:

<https://attacker.example.com>

The destination domain is unrelated to MetroFax and may redirect users to a malicious website.

Indicator: Malicious or deceptive URL

c. Social Engineering Technique

The message uses a common phishing tactic by claiming:

"You have a new fax! Click the attachment to view."

This creates curiosity and encourages immediate action without verification.

Indicator: Social engineering

d. Brand Impersonation

The email claims to originate from MetroFax but does not use an official MetroFax sender domain.

Indicator: Brand impersonation

e. Generic and Unusual Subject Line

The subject line:

"Monday, March 7 2022"

does not clearly indicate the purpose of the email and appears unusual for a legitimate fax notification.

Indicator: Suspicious email formatting

4. Risk Assessment

Assessment Area	Result
Sender Verification	Failed
Domain Trustworthiness	Low
Link Safety	Suspicious
Social Engineering Indicators	Present
Brand Authenticity	Unverified
Overall Classification	Phishing (High Risk)

5. Evidence

```
From: WirelessReceived WirelessReceivedWirelessReceived  
WirelessReceivedWirelessReceived WirelessReceivedWirelessReceived  
WirelessReceived <attack@attacker.example.com>  
Subject: Monday, March 7 2022  
Date: Mon, 7 Mar 2022 17:43:57 +0000  
Message-ID: <CPWP152MB42210B2AAFBE0DF508B1251C9A089CPWP152MB4221LAMP_>  
Content-Language: en-GB  
Content-Type: multipart/alternative;  
boundary="_000_CPWP152MB42210B2AAFBE0DF508B1251C9A089CPWP152MB4221LAMP_"  
To: Undisclosed recipients;;  
Return-Path: attack@attacker.example.com  
MIME-Version: 1.0  
  
--_000_CPWP152MB42210B2AAFBE0DF508B1251C9A089CPWP152MB4221LAMP_--  
Content-Type: text/plain; charset="Windows-1252"  
Content-Transfer-Encoding: quoted-printable  
  
SharePoint  
  
You have a new fax! Click the attachment to view.  
  
Print & Preview Here<https://attacker.example.com/>  
  
Fax Details  
  
Date Received: 2022-03-07 08:35:14 CDT  
Type: Attached in pdf  
Number of Pages: 3  
File Size: 10.5MB
```

Case Study 2 – Microsoft Account Unusual Sign-In Activity

1. Email Overview

Field	Details
Email Type	Phishing Email
Risk Classification	Phishing
Risk Level	High
Claimed Organization	Microsoft
Sender Address	no-reply@access-accsecurity.com
Reply-To Address	sotrecognizd@gmail.com
Subject	Microsoft account unusual sign in activity

2. Email Summary

The email claims that unusual sign-in activity has been detected on a Microsoft account. It attempts to alarm the recipient by reporting a login from an unfamiliar location and encourages immediate action.

The message impersonates Microsoft and uses fear-based language to pressure users into responding without verifying authenticity.

3. Header Analysis

Authentication Check	Result
SPF	Failed / Not Verified
DKIM	Failed / Not Verified
DMARC	Failed / Error
Sender Authenticity	Unverified

4. Phishing Indicators Identified

a. Fake Microsoft Domain

The email claims to be from Microsoft but uses the domain:

access-accsecurity.com

This domain is not owned by Microsoft.

Indicator: Domain impersonation

b. Suspicious Reply-To Address

The email requests reply through:

sotrecognizd@gmail.com

Legitimate Microsoft security notifications do not redirect users to personal Gmail accounts.

Indicator: Reply-To mismatch

c. Authentication Failures

SPF, DKIM, and DMARC checks were not successfully validated.

Indicator: Failed sender authentication

d. Fear-Based Language

The email claims unusual login activity from a foreign location and attempts to create panic.

Indicator: Social engineering / fear tactic

e. Security Alert Impersonation

The email imitates a legitimate account security notification to gain user trust.

Indicator: Brand impersonation

5. Risk Assessment

Assessment Area	Result
Sender Verification	Failed
SPF Validation	Failed
DKIM Validation	Failed
DMARC Validation	Failed
Domain Legitimacy	Suspicious
Social Engineering Indicators	Present
Overall Classification	Phishing (High Risk)

6. Evidence

OriginalChecksum:6DB22130536EEE635A126CD816D1A4DD37580CD58231D7469764E41AA10948F;UpperCasedChecksum;SizeAsReceived:385;Count:12
From: Microsoft account team ,no-reply@access-accsecurity.com
Subject: Microsoft account unusual signin activity
To: phishing@pot
Content-Length: 18708448
Content-Length: 1821750
Date: Fri, 8 Sep 2023 05:47:04 +0000
Reply-To: sotrecognizd@gmail.com
Content-Type: text/html; charset="UTF-8"
Content-Transfer-Encoding: 8bit
X-IncomingHeaderCount: 12
Message-ID:
032672b4-77ca-42f8-a036-9711e91bd1f3@DB8EUR06FT032.eop-eur06.prod.protection.outlook.com
Return-Path: bounce@thculturaldes.co.uk
X-MS-Exchange-Organization-ExpirationStartTime: 08 Sep 2023 05:47:04.2927

Case Study 3 – AccioJob Daily Curated Jobs Email

1. Email Overview

Field	Details
Email Type	Legitimate Email
Risk Classification	Safe
Risk Level	Low
Organization	AccioJob
Sender Address	placement@acciojob-team.org
Subject	daily curated jobs for you

2. Email Summary

The email provides curated job opportunities and career-related recommendations. The content is consistent with recruitment and job placement services offered by AccioJob.

The message contains job listings, application links, company information, and unsubscribe functionality commonly found in legitimate marketing and recruitment emails.

3. Header Analysis

Authentication Check	Result
SPF	Pass
DKIM	Pass
DMARC	Pass
Sender Authenticity	Verified

4. Legitimate Indicators Identified

a. Successful SPF Validation

The email passed SPF authentication, indicating that the sending server is authorized to send emails on behalf of the domain.

Indicator: Legitimate sender authentication

b. Successful DKIM Validation

The email passed DKIM verification, confirming that the message content was not altered during transmission.

Indicator: Message integrity verified

c. Successful DMARC Validation

The email passed DMARC validation, demonstrating alignment between sender identity and authentication records.

Indicator: Domain authenticity verified

d. Consistent Branding

The sender information, content, and branding consistently reference AccioJob services.

Indicator: Brand consistency

e. Presence of Unsubscribe Option

The email includes unsubscribe functionality, which is commonly present in legitimate marketing communications.

Indicator: Compliance with email marketing practices

5. Risk Assessment

Assessment Area	Result
Sender Verification	Passed
SPF Validation	Passed
DKIM Validation	Passed
DMARC Validation	Passed
Domain Legitimacy	Trusted
Social Engineering Indicators	Not Observed
Overall Classification	Safe (Low Risk)

6. Evidence

Google Admin Toolbox: Messageheader

MessageId: 80614863.4457957.177079608128@ip-15-11-01-140.eu-west-1.compute.internal

Created at: 5/26/2026, 12:38:08 AM GMT+5:30 (Delivered after 11 sec)

From: Team Account <placement@accogab-team.org>

To: vkr.freshah3@gmail.com

Subject: daily control jobs for you

DKIM: [pass with domain accogab-team.org](#)

[Link to source](#)

#	Order	From *	To *	Protocol	Time received
1	5 sec	substate.office.com	PH3PR07M99567.40390001.0000.OUTLOOK.COM		5/26/2026, 12:38:13 AM GMT+5:30
2	5 sec	MAOPR01CUB09.outbound.protection.outlook.com	Google mx.google.com	SMTPS	5/26/2026, 12:38:19 AM GMT+5:30
3			Google 2002-a01:6a00:a0b:6010f:9a5:8000	SMTP	5/26/2026, 12:38:19 AM GMT+5:30
4			Google 2002:a01:6a00:a0b:6010f:9a5:8000	SMTP	5/26/2026, 12:38:19 AM GMT+5:30

Common Phishing Indicators

1. Suspicious Sender Address

Attackers frequently use email addresses that imitate legitimate organizations or use domains unrelated to the company they claim to represent.

Example:

support-secure@random-domain.com

Instead of:

support@microsoft.com

2. Generic Greetings

Phishing emails often use generic greetings such as:

- Dear User
- Dear Customer
- Valued Member

Legitimate organizations often address users by their registered name.

3. Urgency and Fear Tactics

Attackers create panic to encourage immediate action.

Examples:

- Your account will be suspended.
- Unusual login activity detected.
- Immediate action required.
- Payment failed.

4. Suspicious Links

The visible text may appear legitimate, but the destination URL may redirect users to a malicious website.

Users should always verify links before clicking.

5. Brand Impersonation

Cybercriminals frequently impersonate trusted organizations such as:

- Microsoft
- Google
- Banks
- Government agencies
- Delivery services

The goal is to gain user trust.

6. Unexpected Attachments

Phishing emails often include unexpected:

- PDF files
- ZIP files
- Office documents
- Executable files

These attachments may contain malware.

7. Authentication Failures

Legitimate organizations commonly implement email authentication standards such as:

- SPF
- DKIM
- DMARC

Authentication failures may indicate spoofing attempts.

8. Requests for Sensitive Information

Legitimate organizations rarely request:

- Passwords
- OTPs
- Banking credentials
- Personal information

through email.

Such requests should be treated as suspicious.

Email Risk Classification Framework

Safe Email (Low Risk)

Characteristics

- Verified sender identity
- SPF, DKIM, and DMARC authentication pass
- Legitimate business purpose
- Trusted domains
- No suspicious links or attachments

User Action

The email may be interacted with normally while maintaining standard security awareness practices.

Example

AccioJob Daily Curated Jobs Email

Suspicious Email (Medium Risk)

Characteristics

- Unknown or unexpected sender
- Limited authentication information
- Unusual wording or formatting
- Unverified links or attachments
- Insufficient evidence to confirm phishing

User Action

Do not interact immediately. Verify authenticity through official communication channels before taking any action.

Phishing Email (High Risk)

Characteristics

- Suspicious sender address
- Domain impersonation
- Authentication failures
- Malicious or deceptive links
- Urgency and fear-based language
- Requests for sensitive information

User Action

Do not click links, open attachments, or provide information. Report the email to the appropriate security team.

Classification Results

Email Sample	Classification	Risk Level
MetroFax Fax Notification	Phishing	High
Microsoft Account Alert	Phishing	High
AccioJob Job Newsletter	Safe	Low

Prevention Guidelines

1. Verify the Sender

Before responding to an email, carefully examine the sender's email address and domain name.

Attackers often use domains that closely resemble legitimate organizations.

Example

Legitimate:

support@microsoft.com

Suspicious:

support@micr0soft-security.com

2. Inspect Links Before Clicking

Hover over links to view the actual destination URL before clicking.

Do not trust link text alone, as attackers may disguise malicious URLs.

3. Be Cautious of Urgent Requests

Treat emails requesting immediate action with caution.

Common phishing messages include:

- Account suspension warnings
- Password reset requests
- Unusual login notifications
- Payment or invoice alerts

4. Never Share Sensitive Information

Do not provide:

- Passwords
- OTPs
- Banking credentials
- Personal information

through email.

Legitimate organizations rarely request such information via email.

5. Verify Requests Through Official Channels

If an email appears suspicious, contact the organization directly through its official website or customer support channel.

Do not rely solely on information provided in the email.

6. Review Email Authentication Indicators

When possible, verify SPF, DKIM, and DMARC authentication results to confirm sender legitimacy.

7. Report Suspicious Emails

Users should immediately report suspicious emails to the organization's security team or IT department.

Early reporting helps prevent wider attacks.

8. Participate in Security Awareness Training

Regular cybersecurity awareness training helps employees recognize evolving phishing techniques and social engineering tactics.

Key Takeaway

Think before you click.

When an email creates urgency, requests sensitive information, or contains suspicious links, verify its authenticity before taking any action.

Employee Do's & Don'ts for Phishing Prevention

✔ DO's

Verify Sender Information

Always check the sender's email address and domain before responding or clicking links.

Inspect Links Carefully

Hover over links to view the destination URL before clicking.

Confirm Suspicious Requests

Verify unexpected requests through official communication channels.

Report Suspicious Emails

Inform the IT or security team immediately when a suspicious email is received.

Use Multi-Factor Authentication (MFA)

Enable MFA wherever possible to provide an additional layer of security.

Stay Updated on Security Awareness

Participate in cybersecurity awareness programs and stay informed about common phishing techniques.

✗ DON'Ts

Do Not Click Unknown Links

Avoid clicking links from unfamiliar or suspicious emails.

Do Not Open Unexpected Attachments

Attachments may contain malware or malicious code.

Do Not Share Sensitive Information

Never disclose passwords, OTPs, banking information, or personal credentials through email.

Do Not Trust Urgent Messages Blindly

Attackers often use urgency and fear to manipulate users into taking immediate action.

Do Not Ignore Security Warnings

Pay attention to browser warnings, email security alerts, and authentication failures.

Do Not Assume Familiar Branding Means Safety

Attackers frequently impersonate trusted organizations such as Microsoft, Google, banks, and delivery services.

Security Reminder

Pause. Verify. Then Act.

Taking a few extra seconds to verify an email can prevent credential theft, financial loss, and unauthorized access to organizational systems.

Conclusion

Phishing attacks continue to be one of the most effective methods used by cybercriminals to compromise user accounts, steal sensitive information, and gain unauthorized access to systems.

This assessment analysed three email samples using header analysis, content inspection, sender verification, and authentication checks. Each email was classified based on identified indicators and associated risk levels.

Key Findings

- Two email samples were identified as phishing emails and classified as High Risk.
- One email sample was identified as a legitimate email and classified as Low Risk.
- Common phishing techniques observed included:
 - Sender spoofing
 - Domain impersonation
 - Suspicious links
 - Social engineering tactics
 - Authentication failures
- Legitimate emails demonstrated successful SPF, DKIM, and DMARC authentication along with consistent sender information.